

AWS DevOps Hands-on Lab Guide

Installing Jenkins on Amazon Linux 2023 using PuTTY (.ppk)

Lab Objective: Launch an EC2 instance, connect with PuTTY, install Java, Git and Jenkins, unlock Jenkins and create the administrator account.

Setting	Value
AMI	Amazon Linux 2023
Instance	t2.micro
Ports	22,8080
User	ec2-user

Step 1 - Login

Login to AWS Console and open EC2 service.

Step 2 - Launch EC2

Click Launch Instance. Name: Jenkins-Server. Choose Amazon Linux 2023. Instance type: t2.micro. Create/select key pair. Enable public IP.

Step 3 - Security Group

Add inbound rules: SSH 22 (My IP), Custom TCP 8080 (Anywhere or My IP for training). Launch the instance and wait until Running.

Step 4 - Convert PEM to PPK

Open PuTTYgen, Load the .pem file, Save Private Key as Jenkins.ppk.

Step 5 - Connect

Open PuTTY. Host: ec2-user@. Port:22. Under Connection > SSH > Auth > Credentials browse to Jenkins.ppk and click Open.

Step 6 - Update

Command: `sudo dnf update -y`

Step 7 - Java

Install: `sudo dnf install java-21-amazon-corretto -y`. Verify using `java -version`.

Step 8 - Git

Install: `sudo dnf install git -y`. Verify using `git --version`.

Step 9 - Jenkins Repository

Create `/etc/yum.repos.d/jenkins.repo` with Jenkins repository. Use `gpgcheck=0` for classroom labs if GPG issues occur.

Step 10 - Import Key

```
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

Step 11 - Refresh Cache

```
sudo dnf clean all then sudo dnf makecache
```

Step 12 - Install Jenkins

```
sudo dnf install jenkins --nogpgcheck -y
```

Step 13 - Start Service

sudo systemctl enable jenkins ; sudo systemctl start jenkins ; sudo systemctl status jenkins. Press q to exit status.

Step 14 - Browser

Copy EC2 Public IPv4 address and browse to http://:8080

Step 15 - Unlock

Run: sudo cat /var/lib/jenkins/secrets/initialAdminPassword and paste password into Jenkins.

Step 16 - Plugins

Select Install Suggested Plugins and wait.

Step 17 - Admin User

Create username, password, full name and email. Save and Finish. Click Start using Jenkins.

Verification

Dashboard should display New Item, Build History, Manage Jenkins, Credentials and People.

Troubleshooting

If port 8080 is inaccessible, verify Security Group, Jenkins service status, and 'sudo ss -tulnp | grep 8080'. If installation fails because of GPG, recreate the repository and use --nogpgcheck for lab purposes.